MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Core, Think Harder and Challenge questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Ages 11-13 (finishing P6)

Math Inquirers P6 Ages 11-13 • Maths + coding bridge. Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three levels. Every child starts with Core, then moves up when they are ready. Challenge questions are the ones to talk about together.
- CORE questions can be started by a child who has not yet learnt the topic, including reading and tracing short programs. Use them for warm-ups and to build confidence.
- THINK HARDER uses the idea taught in the lesson in a new setting: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- CHALLENGE questions are for children who have completed the topic. They need the topic plus an extension: find every way, work backwards, test a case, justify an answer, change the problem or make up your own. This program is inspired by research-based approaches used in gifted education; it is not an official GEP standard and is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Some questions ask you to predict, trace or plan a short program. Children can test their answer in Scratch (or Python) after they have written their prediction on paper.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Code the Pattern (Sequences and the nth Term)	6 + 8 + 9 questions
Lesson 2	Prime Hunters (Primes, Factors and Prime Factorisation)	6 + 8 + 9 questions
Lesson 3	Below Zero (Negative Numbers)	6 + 8 + 9 questions
Lesson 4	Algebra Machines (Expressions and Equations)	6 + 8 + 9 questions
Lesson 5	Every Possible Way (Counting)	6 + 8 + 9 questions
Lesson 6	Fair or Unfair? (Probability)	6 + 8 + 9 questions
Lesson 7	Turtle Geometry (Angles in Polygons)	6 + 8 + 9 questions
Lesson 8	Secret Codes (Clock (Modular) Arithmetic)	6 + 8 + 9 questions
Lesson 9	Data Detectives (Averages and Data)	6 + 8 + 9 questions

Lesson 10	Showcase Challenge (Mixed: Maths and Code)	6 + 8 + 9 questions

Lesson 1: Code the Pattern

Sequences and the nth Term • Week 1

CORE

C1. Write the next two terms: 8, 13, 18, 23, ...

C2. Matchstick triangles in a row. Pattern 1 is 1 triangle (3 sticks). Pattern 2 is 2 triangles side by side (5 sticks). Pattern 3 is 3 triangles (7 sticks). Use matchsticks or draw them. How many sticks does pattern 4 need?

C3. Write out the terms one by one to find the 10th term of 7, 10, 13, 16, ...

C4. Write the next two terms: 40, 36, 32, 28, ...

C5. What does this code say? set n to 10, then repeat 3 [change n by 4], then say n.

C6. A straight fence is 12 m long with a post every 4 m, including a post at each end. Draw it and count the posts. Then work out how many posts a 60 m fence needs.

THINK HARDER

T1. Find the 100th term of 12, 19, 26, 33, ...

T2. Is 200 a term of 2, 9, 16, 23, ...? Explain.

T3. Matchstick houses in a row: 1 house uses 6 sticks, 2 houses use 11, 3 houses use 16. How many houses can you make with 100 sticks?

T4. set n to 7, then repeat 49 [change n by 6], then say n. What does it say? Which term of 7, 13, 19, ... is that? Check with the rule 6n + 1.

T5. The 5th term of a sequence is 23 and the 6th term is 27. The difference is always the same. What is the 1st term? What is the rule?

T6. The 1st term of a sequence is 4 and the 10th term is 40. The difference is always the same. What is the difference?

T7. Write a sequence whose 1st term is 10 and whose 100th term is 1000.

T8. How many terms of 3, 7, 11, 15, ... are less than 500?

CHALLENGE (non-routine: justify, find every way)

X1. Pattern 1 is 1 tile, pattern 2 is a 2 by 2 square of tiles, pattern 3 is a 3 by 3 square. How many tiles are in pattern 10? How many more tiles does pattern 11 need than pattern 10?

X2. Ben counts 6, 11, 16, 21, ... and Mei counts 83, 81, 79, 77, ... They say their terms at the same time. When do they say the same number, and what is it?

X3. The rule for a sequence is 6n − 4. Ben says every term is even. Is he right? Explain.

X4. Find the first term of 5, 8, 11, 14, ... that is bigger than 1000. Which term is it? Then find the first term bigger than 1000 that ends in 0.

X5. A sequence goes 2, 6, 18, 54, ... Is the difference the same each time? What is the rule? What is the 7th term?

X6. What does this code say? set a to 1, set b to 1, then repeat 5 [change a by b, change b by a], then say b.

X7. Make up your own sequence puzzle. Choose a first term and a difference, write the first four terms, then ask: “Is ___ a term?” Pick a number that is NOT a term but is close to one. Swap with a partner and explain how you know.

X8. Which numbers below 60 are in BOTH sequences 2, 5, 8, 11, … and 4, 9, 14, 19, …? Find them all.

X9. What does this code say? set t to 0, set k to 0, then repeat 5 [change k by 1, change t by k], then say t. What would it say with repeat 100?

Lesson 2: Prime Hunters

Primes, Factors and Prime Factorisation • Week 1

CORE

C1. List all the factors of 18. Hint: find pairs of numbers that multiply to 18.

C2. A number is called prime if it has exactly two factors: 1 and itself. For example, 7 is prime (factors 1 and 7) but 8 is not (factors 1, 2, 4, 8). Is 21 prime? Is 23 prime? List the factors of each.

C3. 29 sweets are shared equally among 4 children. How many sweets are left over? (In code, the left-over number is written 29 mod 4.)

C4. Write 30 as a multiplication of as many numbers (bigger than 1) as you can. For example, 30 = 5 × 6 uses two numbers. Can you use three? Can you use four?

C5. A prime number has exactly two factors. Which of these are prime: 21, 23, 25, 27, 29? Hint: try dividing each by 2, 3 and 5.

C6. The primes begin 2, 3, 5, 7, 11, 13, 17, 19. Write 20 as the sum of two of them.

THINK HARDER

T1. How many factors does 64 have? List them.

T2. Find the smallest number that has exactly 6 factors.

T3. Write 100 as the sum of two primes in every possible way.

T4. What does this code say? set count to 0, set d to 1, then repeat 45 [if 45 mod d = 0 then change count by 1; change d by 1], then say count.

T5. Twin primes differ by 2, like 11 and 13. Find all the twin prime pairs below 50.

T6. A number is 2 × 2 × 5 × 7. What is the number? How many factors does it have?

T7. Which number below 30 has the most factors?

T8. A number between 20 and 40 leaves remainder 1 when divided by 2, by 3 and by 4. Find both possible numbers.

CHALLENGE (non-routine: justify, find every way)

X1. What is the largest prime number below 200?

X2. Ali multiplies two different primes and gets 91. Mei multiplies three different primes and gets 105. Which primes did each of them use?

X3. Find the smallest number that can be divided exactly by 2, 3, 4, 5 and 6.

X4. 1001 = 7 × 11 × 13. Use this to write 3003 and 5005 as products of primes.

X5. Find all the numbers between 100 and 200 that have exactly 3 factors.

X6. Which even number below 50 can be written as the sum of two primes in the most ways? How many ways?

X7. Make up a “Who am I?” number puzzle using factors and primes, for example: “I am between 30 and 60 and I have exactly three factors.” Write 2 or 3 clues so that only ONE number fits. Swap with a partner, then check that no other number fits.

X8. A farmer counts her eggs in 2s, 3s, 4s, 5s and 6s, and each time there is 1 egg left over. When she counts them in 7s, none are left over. What is the smallest number of eggs she could have?

X9. 20 lockers are closed. Student 1 opens every locker. Student 2 changes every 2nd locker (open to closed, closed to open). Student 3 changes every 3rd locker, and so on up to student 20. Which lockers are open at the end?

Lesson 3: Below Zero

Negative Numbers • Week 2

CORE

C1. On a thermometer, 0 °C is where water freezes. Colder than that, numbers have a minus sign: −3 °C means 3 degrees below zero. Draw a thermometer or number line and write these in order, coldest first: −4, 2, −10, 0, −1.

C2. It is −3 °C (3 degrees below zero). The temperature rises by 8 degrees. What is it now? Count up on a thermometer drawing.

C3. It is 2 °C. The temperature falls by 7 degrees. What is it now? Count down on a thermometer drawing: 2, 1, 0, then below zero.

C4. Which is colder, −6 °C or −11 °C? By how many degrees? Mark both on a thermometer drawing.

C5. In a building the floors are numbered ..., −2, −1, 0, 1, 2, ... where 0 is the ground floor and −1 is the first floor below it. A lift is at floor −2. It goes up 6 floors. Which floor is it on now? Count the floors one by one.

C6. In Scratch the stage is a number line. A sprite is at x = 30. It runs “change x by −50” (move 50 steps to the left). Where is it now?

THINK HARDER

T1. A sprite starts at x = a. It runs “change x by −8” and then “change x by 5”. It ends at x = −1. What was a? Work backwards from the end, and check by running it forwards.

T2. A diver is at −14 m. A seagull is flying at 9 m. How far apart are they?

T3. Use the numbers −2, 3 and 5 once each to fill the gaps in ____ − ____ − ____. What is the biggest answer you can make? What is the smallest?

T4. City A is 5 degrees colder than city B. City B is 3 degrees warmer than city C. City C is at −4 °C. Find the temperature of B and of A.

T5. At 6 a.m. it was −4 °C. By 2 p.m. it was 7 °C. By 10 p.m. it had fallen 9 degrees. How much did it rise from 6 a.m. to 2 p.m.? What was the temperature at 10 p.m.?

T6. set n to 10, then repeat 5 times: change n by −4. Then say n. What does it say?

T7. What number is exactly halfway between −9 and 3?

T8. A submarine is at −120 m. It rises 45 m, then dives 70 m. Where is it now?

CHALLENGE (non-routine: justify, find every way)

X1. Put −4, −3, −2, −1, 0, 1, 2, 3, 4 in a 3 by 3 grid so that every row, every column and both diagonals add up to 0.

X2. At midnight it was −6 °C. It rose 2 degrees every hour until 7 a.m., then 3 degrees every hour. What was the temperature at 10 a.m.?

X3. set x to 0, then repeat 6 times: change x by 50, change x by −80. Then say x. What does it say?

X4. Which is bigger, −3 − 4 or −3 − (−4)? By how much?

X5. A sprite starts at x = −10. It moves +5, then −3, then +5, then −3, and so on. After how many moves is it first at a number greater than 0? Where is it?

X6. Lift A starts at the surface of a mine (0 m) and goes down 35 m every minute. Lift B starts at −300 m and goes up 25 m every minute. When are they at the same height, and at what height?

X7. Make up your own temperature story with at least three changes (rises and falls) that starts below zero and ends at exactly −7 °C. Write the story, the working and a number-line sketch. Swap with a partner and check each other's.

X8. Two numbers add up to −2. Their difference is 10. What are the two numbers?

X9. In some Singapore buildings there is no level 0. (Q6 on page 2 uses 0 for the ground floor; here, level 1 is the ground floor and B1 is the level below it.) A lift goes from level 4 down to B3. How many floors does it go down? (Count the levels one by one; do not use a number line with 0.)

Lesson 4: Algebra Machines

Expressions and Equations • Week 2

CORE

C1. A number machine multiplies IN by 4 and then adds 1. (We write this OUT = 4n + 1.) What comes OUT when 6 goes IN? Try 2 and 3 first.

C2. A machine changes IN 1, 2, 3, 4 into OUT 3, 5, 7, 9. Look at how OUT goes up each time. What does the machine do? What will come OUT when 10 goes IN?

C3. I think of a number and multiply it by 3. I get 54. What was my number? (Guess and check, or share 54 into 3 equal groups.)

C4. A machine multiplies by 6, then takes away 4. OUT is 38. What was IN? Try some numbers, or undo the steps.

C5. Ali has some stickers. Mei has 5 more than Ali. Fill in the table: if Ali has 3 stickers, Mei has __ and they have __ altogether. If Ali has 7, they have __ altogether. If Ali has 10, they have __. What do you notice about how the total changes?

C6. A Scratch machine runs: ask for IN, then set out to (answer * 3) + 4, then say out. (* means ×.) You type 7. What does it say?

THINK HARDER

T1. IN 1, 2, 3, 4 gives OUT 1, 7, 13, 19. Find the rule. What is OUT when IN is 20?

T2. Solve 9n − 7 = 65.

T3. Machine A does × 2, then + 10. Machine B does + 5, then × 2. Put 7 into both. What do you notice? Explain with n.

T4. I think of a number, divide it by 4, then add 6. I get 15. What is my number?

T5. IN 3 gives OUT 11. IN 7 gives OUT 23. The rule is "× a number, then + a number". Find it.

T6. set n to 2, then repeat 3 times: set n to (n × 2) + 1. Then say n. What does it say?

T7. Solve (n + 6) × 4 = 60.

T8. A taxi charges $4 to start, plus $2 for every km. Write the cost in dollars for n km. How far can you go for $30?

CHALLENGE (non-routine: justify, find every way)

X1. A machine does × 3, then − 4. Which IN number comes out as the SAME number?

X2. Two machines are joined: the OUT of the first goes into the second. Each one does × 2, then + 1. IN 5 gives OUT 23. Which IN gives OUT 63?

X3. Three numbers in a row add up to 87. Call the smallest n. Write the sum with n and find the numbers.

X4. set n to 0, then repeat until (n × 7) + 3 > 100: change n by 1. Then say n. What does it say?

X5. Machine M does × 5, then − 3. Machine N does × 2, then + 15. Which IN gives the same OUT on both machines? What is that OUT?

X6. Which is bigger, 3n + 5 or 3(n + 2)? Does it depend on n?

X7. Find two different machines of the form "× a number, then + or − a number" that both turn IN 5 into OUT 18. Then find the machine of this form that ALSO turns IN 2 into OUT 6. Is there only one?

X8. Mei is n years old. Her mother is 3 times as old. In 12 years, her mother will be twice as old as Mei. How old is Mei now?

X9. Ali's secret machine does × a number, then + or – a number. IN 4 gives OUT 9. IN 9 gives OUT 24. What does IN 20 give? Which IN gives OUT 99?

Lesson 5: Every Possible Way

Counting • Week 3

CORE

C1. A shop sells 3 shirts and 5 pairs of trousers. How many different outfits (1 shirt and 1 pair of trousers) can you make? Draw lines to match each shirt with each pair of trousers.

C2. List every 2-digit number you can make from the cards 4, 6 and 9, using each card at most once.

C3. Four friends meet. Every pair of friends shakes hands once. How many handshakes are there? Act it out, or draw 4 dots and join every pair.

C4. How many times does this say “hi”? repeat 4 { repeat 6 { say hi } } (The inner repeat runs all 6 times for each time round the outer repeat.)

C5. A grid is 3 squares wide and 1 square tall. Draw it. How many routes go from the bottom left corner to the top right corner, going only right or up?

C6. A lock has 2 dials. Each dial shows a digit from 0 to 9. How many different codes are there? Start by counting the codes that begin with 0.

THINK HARDER

T1. A set lunch is 1 main, 1 drink and 1 dessert. There are 4 mains and 3 drinks. There are 60 different set lunches. How many desserts are there?

T2. How many 3-digit numbers can you make from the cards 1, 2, 3 and 4, using each card at most once? How many of them are even?

T3. In a league of 6 teams, every team plays every other team twice (once at home, once away). How many matches are played?

T4. A 4-digit PIN uses only the digits 1, 2 and 3. Digits may repeat. How many PINs are there?

T5. A hexagon has 6 corners. How many diagonals does it have?

T6. How many rectangles of any size can you find in a strip of 4 squares in a row?

T7. 5 children line up for a photo. In how many different orders can they stand? In how many of these is Ali at the front?

T8. How many routes go from one corner of a grid 4 squares wide and 2 squares tall to the opposite corner, going only right or up?

CHALLENGE (non-routine: justify, find every way)

X1. How many different triangles can you make by joining 3 of the 5 corners of a pentagon? What about a hexagon?

X2. How many 3-digit numbers have three different digits? (Remember: a number cannot start with 0.)

X3. How many digits are printed to number the pages of a book from page 1 to page 120?

X4. How many routes go from one corner of a 4 by 4 grid of squares to the opposite corner, going only right or up?

X5. At a party every friend sends a card to every other friend (A to B and B to A are two cards). 90 cards are sent altogether. How many friends are at the party? Test small parties first.

X6. How many squares of any size are there on a 3 by 3 grid of squares?

X7. Make up your own “how many ways” puzzle whose answer is exactly 24. Write the puzzle, show how you counted, and check with a list or a tree. Swap with a partner.

X8. A set of dominoes has one domino for every pair of spot numbers from 0 to 6, including doubles such as 3 | 3. How many dominoes are in the set?

X9. How many 3-digit numbers have digits that add up to 5? (For example, 203.)

Lesson 6: Fair or Unfair?

Probability • Week 3

CORE

C1. Roll a dice. Write down every possible result. How many of them are even? So the chance of an even number is __ out of __. Write it as a fraction.

C2. A bag has 3 red and 7 yellow counters. You pick one without looking. How many counters are there altogether? What is the chance (as a fraction) that you pick yellow?

C3. You flip a coin 200 times. About how many heads do you expect? Will it be exactly that number?

C4. A spinner has 4 equal parts: red, red, blue, green. Write the chance of spinning red as a fraction.

C5. Roll a normal dice. Can you roll a 9? Choose one: impossible, unlikely, likely or certain.

C6. A game: you win if the dice shows 1 or 2; your friend wins if it shows 3, 4, 5 or 6. Write the six dice results and who wins each one. Is the game fair? Why or why not?

THINK HARDER

T1. A spinner has 10 equal parts numbered 1 to 10. What is the probability of spinning a prime number?

T2. Roll two dice and add. What is the probability that the total is 9?

T3. A bag has 5 red counters and some blue counters. The probability of picking red is 1/4. How many blue counters are there?

T4. Ali flips a coin 3 times. What is the probability that he gets heads all 3 times?

T5. One letter is picked at random from the word MATHEMATICS. What is the probability that it is M?

T6. A program rolls a dice 1200 times and counts the numbers bigger than 4. About how many should it count? The program says 391. Is something wrong? Explain.

T7. Roll two dice. What is the probability that at least one of them shows a 6?

T8. Roll a dice. Ali wins on 1 or 2; Ben wins on 3, 4, 5 or 6. Ben scores 1 point for each win. How many points should Ali score for each win to make the game fair?

CHALLENGE (non-routine: justify, find every way)

X1. Three coins are flipped. What is the probability of getting exactly 2 heads?

X2. A number from 1 to 30 is picked at random. What is the probability that it is a multiple of 3 or a multiple of 5?

X3. Roll two dice and multiply the numbers. What is the probability that the answer is odd?

X4. A spinner is spun 1000 times in a program and lands on red 248 times. Is red most likely 1/2, 1/4 or 1/8 of the spinner? Explain.

X5. A bag has 4 red and 6 blue counters. Ali picks a red counter and keeps it. What is the probability that the next counter picked is red?

X6. Ben says: “When you flip two coins you get two heads, two tails, or one of each, so the chance of one of each is 1/3.” Is he right?

X7. Design a FAIR game for two players using one dice, two coins or a spinner. Write the rules and show with a list or table that each player has the same chance of winning. Then change ONE rule to make the game unfair, and say who is favoured.

X8. Mei and Ali each roll one dice. Mei wins if her number is bigger. Ali wins if his number is bigger or the same. Is the game fair? What is the probability that Mei wins?

X9. Spinner A has 3 equal parts: 1, 2, 3. Spinner B has 4 equal parts: 1, 2, 3, 4. Spin both and multiply. Even wins if the answer is even, Odd wins if it is odd. Is the game fair?

Lesson 7: Turtle Geometry

Angles in Polygons • Week 4

CORE

C1. Walk a square: walk forward, turn on the spot by 90° to the right, and repeat, until you are back at the start facing the way you began. You turn 90° at each of the 4 corners. How many degrees did you turn altogether? (A full turn on the spot is 360°.)

C2. A turtle runs: repeat 6 [move 100, turn ↻ 60]. It draws a hexagon. How many degrees does it turn altogether? What do you notice?

C3. A turtle draws an equilateral triangle: repeat 3 [move 100, turn ↻ ___]. It turns the same amount each time and ends up facing the way it started (a full turn is 360°). Guess and check, or draw it with a ruler. What number goes in the gap?

C4. A turtle repeats [move 100, turn ↻ 72] again and again. How many repeats until it first faces the way it started? Add up the turns, 72, 144, ... until you reach a full turn (360). What shape does it draw?

C5. Angles on a straight line add up to 180°. At one corner the turtle turns 50° to go along the next side. What is the inside angle at that corner?

C6. Each turn of a regular hexagon is 60°. A turn and the inside angle at a corner make a straight line (180°). What is each inside angle of a regular hexagon?

THINK HARDER

T1. Each inside angle of a regular polygon is 4 times its turn angle. How many sides does the polygon have?

T2. Each inside angle of a regular polygon is 140°. What do all its inside angles add up to?

T3. Mei's turtle draws a shape with 4 sides. Its first three turns are 70°, 100° and 90°. It comes home facing the same way after going round once. What is its last turn? What is the inside angle at that corner?

T4. A regular polygon has n sides. Write its turn angle and its inside angle using n.

T5. In a regular polygon the turn angle is the same as the inside angle. Which polygon is it?

T6. Which regular polygons with 10 to 20 sides have a whole-number turn angle?

T7. Regular hexagons fit together around a point with no gaps, like a honeycomb. Explain why using angles. Can regular pentagons do the same?

T8. A turtle draws a regular hexagon, then a square, then an equilateral triangle, each once round, ending each shape facing the way it started. How many degrees does it turn altogether?

CHALLENGE (non-routine: justify, find every way)

X1. Ali wants a regular pentagon. He types the inside angle by mistake: repeat 5 [move 100, turn ↻ 108]. Does his shape close? How many repeats would make it close, and what would it draw?

X2. How many different stars with 9 points can a turtle draw using equal moves and equal turns? Give the turns.

X3. The inside angles of a regular polygon add up to 1800°. A second regular polygon has a turn angle exactly half as big as the first one's. How many sides does the second polygon have, and what do all its inside angles add up to?

X4. How many different 12-point stars can a turtle draw with equal moves and equal turns? Give the turn and how many times it goes round.

X5. A turtle runs: repeat 12 [repeat 4 (move 100, turn ↻ 90), turn ↻ 30]. How many full turns does it make altogether?

X6. Zara says: “The more sides a regular polygon has, the bigger each inside angle is, but an inside angle can never reach 180°.” Is she right? Use the turn angle to explain.

X7. Design your own star. Choose the number of points p and how many times it goes round, k, so that the turn angle 360° × k ÷ p is a WHOLE number and p and k share no common factor. Give the turn angle, the point angle (180° – turn) and the code. Explain why your star closes after p repeats.

X8. A square, a hexagon and a 12-sided shape fit exactly around a point: 90° + 120° + 150° = 360°. Find one other set of three regular polygons whose inside angles add up to 360° (hint: the answer uses a square or a triangle).

X9. Try turn 120 and turn 60 with repeat 6 and move 100. Which one draws a hexagon? Why does the other not make a 6-point star?

Lesson 8: Secret Codes

Clock (Modular) Arithmetic • Week 4

CORE

C1. It is 10 o'clock. What time will it be 5 hours later?

C2. A clock has 12 numbers. Start at 12 and move 20 hours forward. Where do you stop? (This is written 20 mod 12: the remainder when 20 is divided by 12.)

C3. What is the remainder when 45 is divided by 7? Hint: count up in 7s.

C4. Today is Monday. What day will it be 10 days from today?

C5. A secret code moves every letter one place along the alphabet: A becomes B, B becomes C, and so on. Code the word DOG. (Write the alphabet out to help you.)

C6. Using the same code (every letter moves one place along), decode KBN. Move each letter one place back.

THINK HARDER

T1. The pattern CODECODECODE... goes on and on. What is the 100th letter?

T2. It is 10 o'clock now. What time will it be 1000 hours from now (on a 12-hour clock)?

T3. Find the smallest number bigger than 10 that leaves a remainder of 3 when it is divided by 4 and also when it is divided by 5.

T4. Decode BYFFI. It was made with a shift of 20.

T5. Today is Monday. What day will it be in 365 days? In 366 days?

T6. Mei codes with a shift of 3. Ben codes with a shift of 29. Do their codes give the same answer? Why?

T7. Work out 47 mod 12, 35 mod 12 and (47 + 35) mod 12. What do you notice?

T8. On a 24-hour clock it is 21:00. What time will it be 50 hours later?

CHALLENGE (non-routine: justify, find every way)

X1. Mei makes a new code: multiply each letter number by 2, then take mod 26. Why can her friend not always decode it?

X2. Ben's code multiplies each letter number by 3, then takes mod 26. Code the word CAB. Do two letters ever get the same code letter?

X3. Crack this Caesar code. Hint: a one-letter word is usually A or I. X ADKT BPIWH

X4. Find the smallest number that leaves a remainder of 2 when it is divided by 3, by 4 or by 5, and is a multiple of 7.

X5. Ali goes to the library every 4th day and Mei every 6th day. They are both there on Day 0, a Monday. On which day of the week do they next meet? And on which day of the week is their 5th meeting after Day 0?

X6. Ali codes a message with a shift of k, then codes the result again with the same shift k. It is the same as one shift of 6. What could k be? Find every answer from 0 to 25.

X7. Write your own secret message of at least 8 letters and code it with a shift of your choice. Write clear instructions telling a friend how to decode it, and explain how a friend could crack it without being told the shift. Swap and decode.

X8. Ali codes a message with a shift of 7. Mei codes Ali's result with another shift. The final message is the same as the original coded with a shift of 2. What shift did Mei use?

X9. Start at 12 on a clock and keep jumping forward 5 hours. Do you land on every number before you get back to 12? What about jumps of 4 hours?

Lesson 9: Data Detectives

Averages and Data • Week 5

CORE

C1. Find the mean of 3, 5, 7 and 9. Hint: share the total equally, or even out the towers.

C2. Find the mean of 10, 0, 8 and 6.

C3. The mean of 5 numbers is 11. What is their total? Think: if the total were shared equally between 5 people, each would get 11.

C4. Four bags have a mean mass of 2.5 kg. What is their total mass?

C5. Find the median (the middle number when they are in order) and the mode (the number that appears most often) of 4, 9, 2, 9, 6.

C6. A Scratch list called marks holds 6, 8 and 7. A loop starts total at 0 and adds every item to total. What is total after the loop? What is the mean?

THINK HARDER

T1. The mean height of 4 children is 150 cm. A fifth child joins the group and the mean height becomes 152 cm. How tall is the fifth child?

T2. Five whole numbers have a mean of 8, a median of 10 and a mode of 12 (12 appears more often than any other number). Find them. Is there more than one answer?

T3. Write 5 whole numbers with a mean of 6 and a mode of 4.

T4. A Scratch list holds 8 numbers with a mean of 12. Then the number 21 is added to the list. What will the mean program say now?

T5. The mean of 7, 11, x and 14 is 10. What is x?

T6. The mean of five numbers is 20. Each number is doubled and then 3 is taken away. What is the new mean?

T7. Ali's mean for 5 spelling tests is 16. His mean for the first 3 tests is 15. What is the mean of his last 2 tests?

T8. Five whole numbers have a mean of 9, a median of 8 and a mode of 8. The smallest is 5 and the largest is 14. Find all five numbers.

CHALLENGE (non-routine: justify, find every way)

X1. Five numbers have a mean of 8. One number is removed and the mean of the other four is 7. (a) Which number was removed? (b) Ali says: "Taking a number out always makes the mean go down." Is he right? Explain, using an example.

X2. A basketball player's mean score for 9 games is 30 points. (a) How many points must she score in the 10th game to raise her mean to 32? (b) The most she has ever scored in one game is 60. What is the highest mean she could have after game 10?

X3. Find ALL the sets of three different counting numbers whose mean is 4 and whose median is 4.

X4. A shop has 10 staff. Nine earn $2000 a month and the manager earns $11 000. Work out the mean, the median and the mode of their pay. The manager wants an advert saying "our staff are well paid": which average should she use? Which average would a worker use to say the pay is low?

X5. The numbers 1 and 5 have a mean of 3. Ali says: “If I add 3 to every number, the mean goes up by 3. If I multiply every number by 3, the mean is multiplied by 3. If I square every number, the mean is squared.” Test each of his three claims. Which are true? Explain with totals.

X6. Five whole numbers have a mean of 6, a median of 5 and only one mode, 2. What is the largest the biggest number can be?

X7. Make up your own “mean mystery” like the Think harder mean puzzles: choose five whole numbers, then write clues (any of mean, median, mode, smallest, largest) that give ONLY your numbers. Swap with a partner. Can they find a different set that fits your clues?

X8. The mean of 1, 2, 3, … , 99 is 50. Explain why without adding them all up. Then plan a short program that checks it.

X9. A group of 8 children has a mean mass of 41 kg. When their tutor joins, the mean mass of all 9 becomes 44 kg. Then one more person joins and the mean of all 10 becomes 43 kg. What is the tutor’s mass? What is the mass of the last person to join?

Lesson 10: Showcase Challenge

Mixed: Maths and Code • Week 5

CORE

C1. Find the 20th term of 5, 8, 11, 14, … (Write the terms out, or look for a shortcut.)

C2. Break 60 into smaller and smaller multiplications until every number is prime (2, 3, 5, 7, 11, …). Write 60 as a product of primes.

C3. The temperature is −3 °C (3 degrees below zero). It falls by 5 degrees. What is it now? Count down on a thermometer drawing.

C4. A machine does × 4, then − 6. What comes out when 5 goes in?

C5. How many different 2-digit numbers can you make with the digit cards 3, 5 and 7 (each card used at most once)?

C6. Find the mean of 12, 15, 9, 20 and 14.

THINK HARDER

T1. A row of squares uses 4, 7, 10, 13, … matchsticks. Which pattern number uses exactly 100 matchsticks?

T2. Find all the prime numbers between 80 and 100.

T3. A regular polygon has an angle of 150° inside each corner. How many sides does it have?

T4. Two dice are rolled and the numbers are added. What is the probability that the total is 10 or more?

T5. Use a Caesar cipher with a shift of 5 to encode the word MATHS.

T6. It is 10 o'clock. What time will a 12-hour clock show 50 hours later?

T7. The mean of 6 numbers is 9. One more number is added and the mean becomes 10. What number was added?

T8. In how many different orders can 4 children stand in a line?

CHALLENGE (non-routine: justify, find every way)

X1. How many of the numbers from 1 to 100 are NOT multiples of 2 or of 3?

X2. A machine does × 3, then − 8. Mei puts a number in and gets the SAME number out. What is it? What if the machine does × 3, then + 8?

X3. How many squares of any size are there on a 4 by 4 grid?

X4. Find ALL the ways to write 50 as the sum of two prime numbers.

X5. A turtle repeats [move 100, turn ↻ 150]. After how many repeats does it first face the way it started? What does it draw, and how many times does it go round?

X6. Three numbers have a mean of 8 and a median of 7. The biggest is 9 more than the smallest. Find the three numbers.

X7. Make up a challenge question that mixes TWO topics from the programme (for example primes and the clock, or sequences and negative numbers). Write the question, the answer and the working. Swap with a partner and solve each other's.

X8. A spinner has the numbers 1 to 8, all equally likely. Ravi spins it twice and adds the numbers. What is the probability that the total is 9? Plan a program that tests it 1000 times.

X9. The nth term of a sequence is 6n − 1. Is 2027 (the year you start Secondary 1) in the sequence? Which is the first term bigger than 2027?

Answer key (tutor only)

Lesson 1: Code the Pattern

C1. 28, 33 (add 5)

C2. 9 sticks (3, 5, 7, 9). Each new triangle adds 2 sticks, because it shares one side with the triangle before it.

C3. 34 (7, 10, 13, 16, 19, 22, 25, 28, 31, 34; the rule 3n + 4 gives 3 × 10 + 4 = 34 too)

C4. 24, 20 (take away 4)

C5. 22 (10 + 4 + 4 + 4)

C6. 4 posts for the 12 m fence (3 gaps). The 60 m fence needs 16 posts: 60 ÷ 4 = 15 gaps, and there is one more post than gaps.

T1. 705. The rule is 7n + 5, and 7 × 100 + 5 = 705.

T2. No. The rule is 7n − 5. The 29th term is 198 and the 30th term is 205, so 200 is skipped.

T3. 19 houses. The rule is 5n + 1: 19 houses use 96 sticks (4 left over) and 20 houses would need 101.

T4. 301, the 50th term (7 + 49 × 6 = 301, and 6 × 50 + 1 = 301). Repeat 49 reaches the 50th term, not the 49th.

T5. 7. Go back 4 jumps of 4: 23 − 16 = 7. The rule is 4n + 3.

T6. 4. From the 1st term to the 10th term is 9 jumps, and 36 ÷ 9 = 4. (The rule is 4n.)

T7. 10, 20, 30, 40, … (rule 10n). There are 99 jumps, and 990 ÷ 99 = 10.

T8. 125. The rule is 4n − 1. The 125th term is 499 and the 126th is 503.

X1. 100 tiles. Pattern 11 has 121 tiles, so 21 more. The differences are 3, 5, 7, 9, ..., so this is NOT a constant-difference sequence.

X2. At the 12th term they both say 61. The gap starts at 77 and closes by 7 each time (Ben +5, Mei −2): 77 ÷ 7 = 11 jumps after the 1st term. Check: 5 × 12 + 1 = 61 and 85 − 2 × 12 = 61.

X3. Yes. 6 × n is always even, and an even number take away 4 is even (2, 8, 14, 20, ...).

X4. 1001, the 333rd term (3n + 2: the 332nd term is 998). The first one bigger than 1000 that ends in 0 is 1010, the 336th term (1001, 1004, 1007, 1010).

X5. No: the differences are 4, 12, 36. Each term is 3 times the one before. The 7th term is 1458 (54, 162, 486, 1458).

X6. 144. a and b leapfrog: 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 (the Fibonacci numbers).

X7. Open. Example: 6, 11, 16, 21, … (rule 5n + 1). Is 100 a term? No: 100 − 1 = 99 is not a multiple of 5 (the 19th term is 96 and the 20th is 101). Look for a reason from the rule or from counting jumps, not just a list.

X8. 14, 29, 44 and 59. One more than each of them is a multiple of both 3 and 5, so a multiple of 15 (15, 30, 45, 60).

X9. 15 (1 + 2 + 3 + 4 + 5). With repeat 100 it says 5050: pair 1 + 100, 2 + 99, … to make 50 pairs of 101.

Lesson 2: Prime Hunters

C1. 1, 2, 3, 6, 9, 18

C2. 21 is not prime (factors 1, 3, 7, 21). 23 is prime (its only factors are 1 and 23).

C3. 1 (4 × 7 = 28, so 1 is left over: 29 mod 4 = 1)

C4. 30 = 2 × 3 × 5 uses three numbers, and you cannot use four: none of 2, 3 and 5 can be split again. (These are the prime factors of 30.)

C5. 23 and 29

C6. 3 + 17 or 7 + 13

T1. 7: 1, 2, 4, 8, 16, 32, 64 (8 × 8 counts once).

T2. 12 (1, 2, 3, 4, 6, 12). Every number below 12 has fewer than 6 factors.

T3. 6 ways: 3 + 97, 11 + 89, 17 + 83, 29 + 71, 41 + 59, 47 + 53.

T4. 6. It counts the factors of 45: 1, 3, 5, 9, 15, 45.

T5. 6 pairs: 3 and 5, 5 and 7, 11 and 13, 17 and 19, 29 and 31, 41 and 43.

T6. 140. It has 12 factors: 1, 2, 4, 5, 7, 10, 14, 20, 28, 35, 70, 140.

T7. 24, with 8 factors (1, 2, 3, 4, 6, 8, 12, 24).

T8. 25 and 37. Each is 1 more than a multiple of 12.

X1. 199. Test 2, 3, 5, 7, 11 and 13 (14 × 14 = 196, 15 × 15 = 225): none of them divides 199.

X2. Ali: 7 × 13. Mei: 3 × 5 × 7.

X3. 60. It needs 2 × 2 (for 4), a 3 and a 5: 2 × 2 × 3 × 5 = 60.

X4. 3003 = 3 × 7 × 11 × 13 and 5005 = 5 × 7 × 11 × 13.

X5. 121 and 169 (11 × 11 and 13 × 13). 144 = 12 × 12 does not work because 12 is not prime.

X6. 48, in 5 ways: 5 + 43, 7 + 41, 11 + 37, 17 + 31, 19 + 29.

X7. Open. Example: "I am between 30 and 60 and I have exactly 3 factors" fits only 49 (7 × 7). The numbers with exactly 3 factors are the squares of primes: 4, 9, 25, 49, 121, … Check a puzzle by listing every number that fits the clues.

X8. 301. It must be 1 more than a multiple of 60: 61, 121, 181, 241, 301, … The first of these in the 7 times table is 301 = 7 × 43.

X9. 1, 4, 9 and 16. Locker n is changed once for each factor of n. Only square numbers have an odd number of factors, so only they end up open.

Lesson 3: Below Zero

C1. −10, −4, −1, 0, 2

C2. 5 °C

C3. −5 °C

C4. −11 °C, by 5 degrees

C5. Floor 4 (−2 + 6 = 4)

C6. x = −20

T1. 2 (work backwards: −1 − 5 = −6, then −6 + 8 = 2; check: 2 − 8 = −6, −6 + 5 = −1)

T2. 23 m (14 up to sea level, then 9 more)

T3. Biggest 4 (5 − 3 − (−2) or 5 − (−2) − 3); smallest −10 (−2 − 3 − 5). Try all 6 orders: the answers are 4, 4, 0, 0, −10, −10.

T4. B is −1 °C (−4 + 3) and A is −6 °C (−1 − 5).

T5. It rose 11 degrees; at 10 p.m. it was −2 °C (7 − 9).

T6. −10 (10 − 20)

T7. −3 (the gap is 12; −9 + 6 = −3 and 3 − 6 = −3)

T8. −145 m (−120 + 45 = −75, −75 − 70 = −145)

X1. One answer: top row −3, 2, 1; middle row 4, 0, −4; bottom row −1, −2, 3. 0 must be in the middle and 4 and −4 cannot be in corners; the other 7 answers are turns and flips of this one.

X2. 17 °C (−6 + 7 × 2 = 8 at 7 a.m., then 8 + 3 × 3 = 17)

X3. −180 (each repeat moves −30; 6 × −30 = −180)

X4. −3 − (−4) = 1 is bigger than −3 − 4 = −7, by 8.

X5. After 7 moves, at x = 1 (−5, −8, −3, −6, −1, −4, 1).

X6. After 5 minutes, at −175 m (the gap of 300 m closes by 60 m each minute).

X7. Open. Example: starts at −2 °C, falls 6 degrees (−8), rises 3 degrees (−5), falls 2 degrees (−7). Check that the story really ends at −7 and that each change is shown on a number line.

X8. 4 and −6 (4 + (−6) = −2 and 4 − (−6) = 10)

X9. 6 floors (4, 3, 2, 1, B1, B2, B3: six moves down). A number line with 0 would give 7, because it counts a level 0 that these buildings do not have.

Lesson 4: Algebra Machines

C1. 25 (6 × 4 = 24, then + 1)

C2. It doubles IN and then adds 1 (OUT = 2n + 1). IN 10 gives 21.

C3. 18

C4. 7 (38 + 4 = 42, 42 ÷ 6 = 7)

C5. Ali 3: Mei 8, total 11. Ali 7: Mei 12, total 19. Ali 10: Mei 15, total 25. Each extra sticker Ali has adds 2 to the total (one for Ali and one for Mei).

C6. 25

T1. OUT = 6n − 5; IN 20 gives 115.

T2. n = 8 (65 + 7 = 72, 72 ÷ 9 = 8)

T3. Both give 24. (n + 5) × 2 = 2n + 10, so they always give the same OUT.

T4. 36 (15 − 6 = 9, 9 × 4 = 36)

T5. OUT = 3n + 2 (IN up 4, OUT up 12, so × 3; 9 + 2 = 11)

T6. 23 (2 → 5 → 11 → 23)

T7. n = 9 (60 ÷ 4 = 15, 15 − 6 = 9)

T8. 2n + 4 dollars; 13 km (30 − 4 = 26, 26 ÷ 2 = 13)

X1. 2 (2 × 3 − 4 = 2; 3n − 4 = n)

X2. 15 (63 → 31 → 15; check 15 → 31 → 63)

X3. n + (n + 1) + (n + 2) = 3n + 3 = 87, so n = 28: the numbers are 28, 29, 30.

X4. 14 (13 gives 94, 14 gives 101)

X5. IN 6, OUT 27 (5n − 3 = 2n + 15, so 3n = 18)

X6. 3(n + 2) = 3n + 6 is always 1 bigger; it does not depend on n.

X7. Many machines turn 5 into 18: × 3 then + 3, × 2 then + 8, × 1 then + 13, × 4 then − 2, ... Only × 4 then − 2 also turns 2 into 6 (4 × 2 − 2 = 6). Two pairs of IN and OUT fix the rule: 5a + b = 18 and 2a + b = 6 give 3a = 12, so a = 4 and b = −2.

X8. 12 (3n + 12 = 2(n + 12), so n = 12; mother 36; in 12 years 24 and 48)

X9. The rule is 3n − 3. IN 20 gives 57. OUT 99 comes from IN 34.

Lesson 5: Every Possible Way

C1. 15 (3 × 5)

C2. 6 numbers: 46, 49, 64, 69, 94, 96

C3. 6 (3 + 2 + 1)

C4. 24 (4 × 6)

C5. 4 (go up at one of the 4 lines across)

C6. 100 (10 × 10: 00 to 99; 10 codes begin with 0 and there are 10 first digits)

T1. 5 (4 × 3 = 12 and 60 ÷ 12 = 5)

T2. 24 (4 × 3 × 2); 12 are even (the last card is 2 or 4: 2 choices, then 3 × 2 for the others)

T3. 30 (6 × 5: each ordered pair is one home match)

T4. 81 (3 × 3 × 3 × 3)

T5. 9 (6 corners make 6 × 5 ÷ 2 = 15 lines; 6 of them are sides, so 9 are diagonals)

T6. 10 (4 of length 1, 3 of length 2, 2 of length 3, 1 of length 4)

T7. 120 (5 × 4 × 3 × 2 × 1); 24 with Ali at the front (4 × 3 × 2 × 1)

T8. 15

X1. 10 for the pentagon (list them, or 5 × 4 × 3 ÷ 6); 20 for the hexagon (6 × 5 × 4 ÷ 6).

X2. 648 (9 choices for the first digit, 9 for the second because 0 is allowed but the first digit is used, 8 for the third: 9 × 9 × 8)

X3. 252 (9 one-digit + 90 × 2 + 21 × 3 = 9 + 180 + 63)

X4. 70 (add as you go: the corner numbers on the diagonal row are 1, 4, 10, 20, 35, then 70)

X5. 10 friends (10 × 9 = 90). For 4 friends it is 4 × 3 = 12, for 5 friends 5 × 4 = 20, and so on.

X6. 14 (9 small, 4 of size 2 by 2, 1 of size 3 by 3)

X7. Open. Examples: 4 children in a line (4 × 3 × 2 × 1 = 24); 2 shirts × 3 shorts × 4 caps = 24 outfits; 3-digit numbers from the cards 1, 2, 3, 4 using each card at most once (4 × 3 × 2 = 24). Check that every case is counted once.

X8. 28 (7 doubles, plus 6 + 5 + 4 + 3 + 2 + 1 = 21 with two different numbers)

X9. 15 (first digit 1: 5 numbers, 2: 4, 3: 3, 4: 2, 5: 1; they are 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500)

Lesson 6: Fair or Unfair?

C1. 3 out of 6 = 1/2 (2, 4 or 6)

C2. 10 counters; 7/10

C3. About 100 (half of 200), but probably not exactly 100

C4. 2/4 = 1/2

C5. Impossible: the probability is 0 (a dice only shows 1 to 6)

C6. You win on 2 of the 6 results and your friend on 4 of the 6, so it is not fair (2/6 = 1/3 against 4/6 = 2/3).

T1. 4/10 = 2/5 (2, 3, 5, 7)

T2. 4/36 = 1/9 (3+6, 4+5, 5+4, 6+3)

T3. 15 (5 out of 20 is 1/4)

T4. 1/8 (8 equally likely outcomes: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT)

T5. 2/11

T6. About 400 (2 of the 6 results are bigger than 4, and 1/3 × 1200 = 400). 391 is close, so nothing is wrong: a random simulation is never exactly the expected number.

T7. 11/36 (6 with the red die, 6 with the blue die, minus the 6 + 6 counted twice: 6 + 6 − 1 = 11)

T8. 2 points (Ali wins 1/3 of the time and Ben 2/3, so Ali needs twice as many points per win)

X1. 3/8 (HHT, HTH, THH out of 8 outcomes)

X2. 14/30 = 7/15 (10 multiples of 3, 6 of 5, and 15 and 30 are counted twice: 10 + 6 − 2 = 14)

X3. 9/36 = 1/4 (both dice must be odd: 3 × 3 = 9 ways)

X4. 1/4: we would expect about 250 reds, and 248 is close. 1/2 would give about 500 and 1/8 about 125.

X5. 3/9 = 1/3 (3 red left out of 9)

X6. No: HH, HT, TH, TT are the 4 equally likely outcomes, and one of each happens 2 ways: 2/4 = 1/2.

X7. Open. Example: roll one dice; Player A wins on 1, 2 or 3 and Player B wins on 4, 5 or 6 (3/6 each). Change: A also wins on 4. Now A has 4/6 and B has 2/6, so A is favoured. Check that every outcome is listed and the winning outcomes add up to the whole.

X8. Not fair. Mei wins 15/36 = 5/12; Ali wins 21/36 = 7/12 (15 bigger + 6 the same). Fair version: the same number means roll again.

X9. Not fair. There are 3 × 4 = 12 outcomes; the answer is odd only when both are odd: 2 × 2 = 4 ways. P(Odd) = 4/12 = 1/3, P(Even) = 8/12 = 2/3.

Lesson 7: Turtle Geometry

C1. 360° (4 × 90°): one full turn

C2. 360° (6 × 60°). The same total as the square: one full turn.

C3. 120 (3 × 120 = 360)

C4. 5 repeats (5 × 72 = 360). It draws a regular pentagon.

C5. 130° (180° − 50°)

C6. 120° (180° − 60°)

T1. 10 (turn + 4 × turn = 180°, so the turn is 36° and 360 ÷ 36 = 10; check: inside 144° = 4 × 36°)

T2. 1260° (turn 40°, so 9 sides; 9 × 140° = 1260°; check (9 − 2) × 180° = 1260°)

T3. Last turn 100° (360 − 70 − 100 − 90); inside angle 80° (180 − 100)

T4. Turn = 360 ÷ n; inside angle = 180 − 360 ÷ n

T5. A square (turn = inside = 90°, because they add up to 180°)

T6. 10, 12, 15, 18 and 20 sides (turns 36°, 30°, 24°, 20°, 18°); 360 cannot be shared exactly into 11, 13, 14, 16, 17 or 19 equal turns

T7. Three hexagon angles make 3 × 120° = 360°. Pentagons cannot: 3 × 108° = 324° leaves a gap and 4 × 108° = 432° overlaps.

T8. 1080° (360° for each shape: 3 × 360°)

X1. No: 5 × 108° = 540° is not a whole number of turns. 108 × 10 = 1080 = 3 × 360, so 10 repeats close it: a 10-point star that goes round 3 times (each point 72°).

X2. 2: turn 80° (round 2 times) and turn 160° (round 4 times). Turn 120° (3 times round) only draws a triangle 3 times; turns of 200°, 280° … give the same stars turned the other way.

X3. First polygon: (n − 2) × 180° = 1800°, so n = 12 and its turn is 30°. Second: turn 15°, so 360 ÷ 15 = 24 sides. Its inside angles add up to (24 − 2) × 180° = 3960° (or 24 × 165°).

X4. Only 1: turn 150°, going round 5 times (5 × 360 ÷ 12). Turns 60°, 90° and 120° (round 2, 3, 4 times) just redraw a hexagon, square or triangle, and 180° goes back and forth on a line.

X5. 13 (12 × (360° + 30°) = 4680° = 13 × 360°)

X6. Yes. The turn is 360° ÷ n. It gets smaller as n grows but is never 0, so the inside angle 180° − 360° ÷ n gets bigger and always stays below 180° (180° would be a straight line, which needs a turn of 0°). For example 5 sides give 108°, 12 sides give 150° and 60 sides give 174°.

X7. Open. Example: 8 points going round 3 times: turn = 360 × 3 ÷ 8 = 135°, point angle 45°; code repeat 8 [move 100, turn ↻ 135]. After 8 repeats it has turned 8 × 135° = 1080° = 3 full turns, so it faces the way it started and the star closes. If p and k share a factor, the star closes sooner and is a smaller shape repeated (9 points going round 3 times is just a triangle).

X8. Square + octagon + octagon (90° + 135° + 135°) or triangle + 12-gon + 12-gon (60° + 150° + 150°).

X9. Turn 60 draws the hexagon (6 × 60° = 360°, one full turn). Turn 120 draws a triangle twice round (6 × 120° = 720°, two full turns), and it only visits 3 corners, not 6.

Lesson 8: Secret Codes

C1. 3 o'clock (15 − 12 = 3)

C2. 8 o'clock (20 − 12 = 8, so 20 mod 12 = 8)

C3. 3 (45 = 6 × 7 + 3)

C4. Thursday (10 mod 7 = 3; Monday + 3 days)

C5. EPH

C6. JAM

T1. E (100 mod 4 = 0, so the 100th letter is the 4th letter of CODE)

T2. 2 o'clock (1000 mod 12 = 4, and 10 + 4 = 14, 14 − 12 = 2)

T3. 23 (3 more than a multiple of 20: 23 = 5 × 4 + 3 = 4 × 5 + 3)

T4. HELLO (B = 1: 1 − 20 = −19, + 26 = 7 = H; Y = 24 − 20 = 4 = E; F = 5 − 20 = −15, + 26 = 11 = L; I = 8 − 20 = −12, + 26 = 14 = O)

T5. Tuesday and Wednesday (365 = 52 × 7 + 1, 366 = 52 × 7 + 2)

T6. Yes: 29 mod 26 = 3, so a shift of 29 is a full turn (26) plus 3.

T7. 11, 11 and 10. Adding the remainders and taking mod 12 again gives the same answer: (11 + 11) mod 12 = 22 mod 12 = 10.

T8. 23:00 (50 mod 24 = 2, and 21 + 2 = 23)

X1. Two letters can give the same code letter: A (0) and N (13) both give 0 = A (13 × 2 = 26). So every code letter could stand for two different letters.

X2. GAD (C = 2 gives 6 = G, A = 0 gives 0 = A, B = 1 gives 3 = D). No: all 26 letters get different code letters, because 3 and 26 have no common factor.

X3. I LOVE MATHS (X = 23 is I = 8, so the shift is 15)

X4. 182 (it is 2 more than a multiple of 60: 62, 122, 182, and 182 = 7 × 26). The smaller numbers 2, 62 and 122 are not multiples of 7.

X5. They next meet on Day 12 (the smallest number in both the 4 and 6 times tables), and 12 mod 7 = 5, so it is a Saturday. Their 5th meeting after Day 0 is on Day 60, and 60 mod 7 = 4, so it is a Friday.

X6. 3 or 16 (3 + 3 = 6; 16 + 16 = 32 and 32 mod 26 = 6)

X7. Open. Check the code by decoding it, and that the cracking idea is sensible: try every shift from 1 to 25, or use the commonest letter (often E) or a short word (A or I). Example: MEET AT SIX with a shift of 3 is PHHW DW VLA.

X8. 21 (7 + 21 = 28, and 28 mod 26 = 2)

X9. Jumps of 5: yes, all 12 numbers (5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 12). Jumps of 4: no, only 4, 8 and 12, because 4 shares a factor with 12.

Lesson 9: Data Detectives

C1. 6 (24 ÷ 4)

C2. 6 (24 ÷ 4: the 0 counts as a value)

C3. 55

C4. 10 kg

C5. In order: 2, 4, 6, 9, 9. Median 6, mode 9.

C6. Total 21, mean 7

T1. 160 cm (5 × 152 = 760, 4 × 150 = 600, 760 − 600 = 160)

T2. Three answers: 0, 6, 10, 12, 12 or 1, 5, 10, 12, 12 or 2, 4, 10, 12, 12. Median 10 is the middle (3rd) number; the mode 12 must appear twice, so the 4th and 5th numbers are 12; the total is 5 × 8 = 40, so the first two add up to 40 − 10 − 24 = 6. (Using 3 and 3 would make 3 a second mode.)

T3. Open, e.g. 4, 4, 5, 8, 9 (total 30, and 4 appears most often). Check the total is 30.

T4. 13 (8 × 12 = 96, 96 + 21 = 117, 117 ÷ 9 = 13)

T5. 8 (4 × 10 = 40, 40 − 7 − 11 − 14 = 8)

T6. 37 (20 × 2 − 3)

T7. 17.5 (5 × 16 = 80, 3 × 15 = 45, 80 − 45 = 35, 35 ÷ 2 = 17.5)

T8. 5, 8, 8, 10, 14 (total 45; the middle one is 8; 8 must appear twice, and it can only be the 2nd number, so the 4th is 45 − 5 − 8 − 8 − 14 = 10)

X1. (a) 12 (5 × 8 = 40, 4 × 7 = 28, 40 − 28 = 12). (b) No. Removing a number above the mean (like 12) lowers the mean, but removing a number below the mean raises it. Example: 4, 6, 8, 10, 12 has mean 8; remove 4 and 6, 8, 10, 12 has mean 9.

X2. (a) 50 (10 × 32 = 320, 9 × 30 = 270, 320 − 270 = 50). (b) 33: 270 + 60 = 330 and 330 ÷ 10 = 33.

X3. 3 sets: 1, 4, 7; 2, 4, 6; 3, 4, 5 (the total is 12, the middle is 4, so the other two add to 8 and one is less than 4)

X4. Mean = (9 × 2000 + 11 000) ÷ 10 = $2900; median = $2000; mode = $2000. The manager would use the mean ($2900) because one big pay lifts it. A worker would use the median or the mode ($2000), which shows what most people earn.

X5. Add 3: 4 and 8 have mean 6 = 3 + 3, true. Multiply by 3: 3 and 15 have mean 9 = 3 × 3, true. Square: 1 and 25 have mean 13, not 9, so false. Adding 3 to each of n numbers adds 3n to the total, so the mean goes up by 3; multiplying every number by 3 multiplies the total by 3.

X6. 15 (2, 2, 5, 6, 15: the total is 30, so the last two make 21; the 4th must be at least 6, or 5 would tie with 2 as a mode)

X7. Open. Example clues: mean 5, median 4, mode 4, smallest 2, largest 9. The total is 25, so the other two numbers add to 25 − 2 − 4 − 9 = 10; only 4 and 6 fit (4 must appear twice to be the mode), giving 2, 4, 4, 6, 9. Check that the clues really give one set.

X8. Pair them: 1 + 99, 2 + 98, … , 49 + 51 each make 100, with 50 left in the middle. Every pair balances around 50, so the mean is 50 (total 4950 ÷ 99 = 50). Program: set total to 0, set n to 1, repeat 99 (change total by n, change n by 1), say total ÷ 99.

X9. Tutor: 68 kg (9 × 44 = 396, 8 × 41 = 328, 396 − 328 = 68). Last person: 34 kg (10 × 43 = 430, 430 − 396 = 34).

Lesson 10: Showcase Challenge

C1. 62 (the nth term is 3n + 2, so 3 × 20 + 2)

C2. 2 × 2 × 3 × 5

C3. −8 °C

C4. 14

C5. 6 (35, 37, 53, 57, 73, 75)

C6. 14 (70 ÷ 5)

T1. Pattern 33 (3n + 1 = 100)

T2. 83, 89 and 97

T3. 12 (the turn at each corner is 180° − 150° = 30°, and 360 ÷ 30 = 12)

T4. 6/36 = 1/6 (10: 3 ways, 11: 2 ways, 12: 1 way)

T5. RFYMX

T6. 12 o'clock (10 + 50 = 60, and 60 is a multiple of 12)

T7. 16 (7 × 10 − 6 × 9)

T8. 24 (4 × 3 × 2 × 1)

X1. 33 (multiples of 2: 50, of 3: 33, of both (6): 16; 100 − (50 + 33 − 16) = 33)

X2. 4 (4 × 3 − 8 = 4). With + 8 it is −4 (−4 × 3 + 8 = −4).

X3. 30 (16 + 9 + 4 + 1)

X4. 4 ways: 3 + 47, 7 + 43, 13 + 37, 19 + 31

X5. 12 repeats (12 × 150° = 1800° = 5 × 360°, and no smaller number of repeats gives a multiple of 360°). It draws a 12-point star going round 5 times.

X6. 4, 7 and 13 (the total is 24, so the smallest and biggest add to 17 and differ by 9)

X7. Open. Example: "The nth term of a sequence is 7n − 3. Which is the first term that is a multiple of 5?" The terms are 4, 11, 18, 25, so it is the 4th term, 25. Check that the answer is correct and that both topics are really needed.

X8. 8/64 = 1/8 (1 + 8, 2 + 7, … , 8 + 1), so about 125 in 1000. Program: set nines to 0; repeat 1000 (if pick random 1 to 8 + pick random 1 to 8 = 9 then change nines by 1); say nines.

X9. Yes: it is the 338th term ($6 \times 338 - 1 = 2027$). The first term bigger than 2027 is the 339th, 2033.